DuPont™ Zytel® 105F BK010

NYLON RESIN

Product Information

Common features of Zytel® nylon resin include mechanical and physical properties such as high mechanical strength, excellent balance of stiffness and toughness, good high temperature performance, good electrical and flammability properties, good abrasion and chemical resistance. In addition, Zytel® nylon resins are available in different modified and reinforced grades to create a wide range of products with tailored properties for specific processes and end-uses. Zytel® nylon resin, including most flame retardant grades, offer the ability to be coloured.

The good melt stability of Zytel® nylon resin normally enables the recycling of properly handled production waste. If recycling is not possible, DuPont recommends, as the preferred option, incineration with energy recovery (~31kJ/g of base polymer) in appropriately equipped installations. For disposal, local regulations have to be observed.

Zytel® nylon resin typically is used in demanding applications in the automotive, furniture, domestic appliances, sporting goods and construction industry.

Zytel® 105F BK010 is a lubricated, fast cycling, weather resistant polyamide 66 resin. Zytel® 105F BK010 contains finely dispersed carbon black.

General information	Value	Unit	Test Standard
Resin Identification	PA66	-	-
Part Marking Code	>PA66<	-	ISO 11469
Rheological properties	**dry / cond**	**Unit**	**Test Standard**
Viscosity number	145 / *	cm^3/g	ISO 307, 1157, 1628
Moulding shrinkage, parallel	1.3 / *	%	ISO 294-4, 2577
Moulding shrinkage, normal	1.3 / *	%	ISO 294-4, 2577
Mechanical properties	**dry / cond**	**Unit**	**Test Standard**
Tensile Modulus	3200 / 1500	MPa	ISO 527-1/-2
Yield stress	85 / 60	MPa	ISO 527-1/-2
Yield strain	4.3 / 25	%	ISO 527-1/-2
Nominal strain at break	24 / >50	%	ISO 527-1/-2
Tensile creep modulus			ISO 899-1
1h	* / 1340	MPa	
1000h	* / 600	MPa	
Charpy impact strength			ISO 179/1eU
23°C	45 / N	kJ/m^2	
-30°C	55 / 55	kJ/m^2	
Charpy notched impact strength			ISO 179/1eA
23°C	6 / 15	kJ/m^2	
-30°C	4 / 3	kJ/m^2	
Izod notched impact strength			ISO 180/1A
23°C	5 / 12	kJ/m^2	
-30°C	4 / 3	kJ/m^2	
Thermal properties	**dry / cond**	**Unit**	**Test Standard**
Melting temperature, 10°C/min	263 / *	°C	ISO 11357-1/-3
Glass transition temperature, 10°C/min	60 / -	°C	ISO 11357-1/-2
Temp. of deflection under load			ISO 75-1/-2
1.8 MPa	70 / *	°C	
0.45 MPa	205 / *	°C	
Vicat softening temperature, 50°C/h, 50N	240 / *	°C	ISO 306
Coeff. of linear therm. expansion, parallel	100 / *	E-6/K	ISO 11359-1/-2
Coeff. of linear therm. expansion, normal	110 / *	E-6/K	ISO 11359-1/-2
RTI, electrical			UL 746B
0.8mm	125 / *	°C	
1.5mm	125 / *	°C	
3mm	125	°C	
6mm	125	°C	

To find out more, visit DuPont Performance Polymers or contact nearest DuPont location.

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RTI, impact			UL 746B
0.8mm	65	°C	
1.5mm	75 / *	°C	
3mm	75	°C	
6mm	75	°C	
RTI, strength			UL 746B
0.8mm	65	°C	
1.5mm	85 / *	°C	
3mm	85	°C	
6mm	85	°C	
Flammability	**dry / cond**	**Unit**	**Test Standard**
Burning Behav. at 1.5mm nom. thickn.	V-2 / *	class	IEC 60695-11-10
Thickness tested	1.5 / *	mm	IEC 60695-11-10
UL recognition	UL / *	-	-
Burning Behav. at thickness h	V-2 / *	class	IEC 60695-11-10
Thickness tested	0.7 / *	mm	IEC 60695-11-10
Oxygen index	27 / *	%	ISO 4589-1/-2
Flammability			IEC 60695-11-10
3.0mm	V-2 / *	-	
6.0mm	V-2 / *	-	
Electrical properties	**dry / cond**	**Unit**	**Test Standard**
Relative permittivity, 1MHz	3.6 / 4.6	-	IEC 60250
Dissipation factor, 1MHz	300 / 600	E-4	IEC 60250
Comparative tracking index	600 / -	-	IEC 60112
Other properties	**dry / cond**	**Unit**	**Test Standard**
Humidity absorption, 2mm	2.7 / *	%	Sim. to ISO 62
Water absorption, 2mm	8.5 / *	%	Sim. to ISO 62
Density	1140 / -	kg/m³	ISO 1183
VDA Properties	**Value**	**Unit**	**Test Standard**
Burning rate, Thickness 1 mm	0[1]	mm/min	ISO 3795 (FMVSS 302)
Emission of organic compounds	6.6	µgC/g	VDA 277
Odour	2.5	class	VDA 270

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Characteristics			
Processing	• Injection Moulding		
Delivery form	• Pellets		
Additives	• Lubricants	• Release agent	
Special characteristics	• Light stabilised or stable to light	• U.V. stabilised or stable to weather	
Regional Availability	• North America • Europe	• Asia Pacific • South and Central America	• Near East/Africa • Global

Processing Texts

Injection molding

PREPROCESSING

Drying recommended = Yes, if moisture content of resin exceeds recommended level
Drying temperature = 80°C
Drying time, dehumidified dryer = 2-4 h
Processing moisture content = <0.2 %

PROCESSING

Melt temperature optimum = 290°C
Melt temperature range = 280-300°C

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Mould temperature optimum = 70°C
Mould temperature range = 50-90°C
Maximum Screw tangential Speed : 0.4 m/s
Flow front speed : 300 mm/s
Hold pressure optimum : 85 MPa
Hold pressure range : 50-100 MPa
Back pressure : low
Hold pressure time : 4 s/mm
Maximum hold-up time : 15 min

Note: 103HSL contains less lubricant than 103FHS and 103FHSA.
It can then lead to more ejection problems.

Contact DuPont for Material Safety Data Sheet, general guides and/or additional information about ventilation, handling, purging, drying, etc. ISO Mechanical properties measured at 4.0mm (Hytrel® measured at 2 mm), IEC Electrical properties measured at 2.0mm, all ASTM properties measured at 3.2mm, and test temperatures are 23°C unless otherwise stated.

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